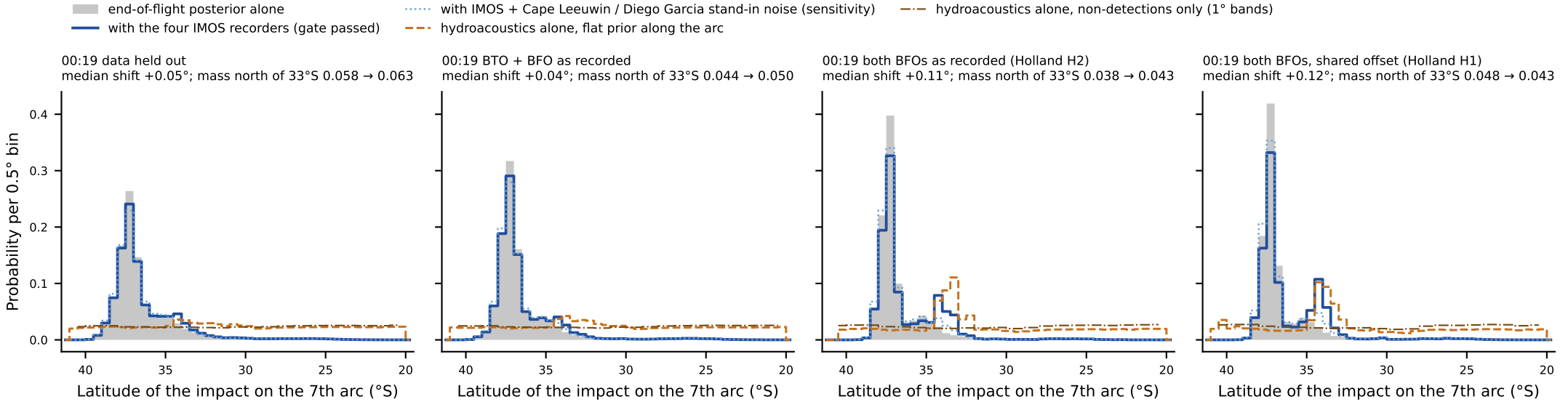
At dive probability 0.5, adding the hydroacoustic result moves the end-of-flight latitude distribution by 0.04–0.12°;
the bumps near 33–36°S come from two time-matched Portland candidates that are too loud to be the aircraft (amplitude-blind matching)



Plain-language note: Grey: where the flight simulation (run J) puts the impact, by latitude. Blue: the same after multiplying by the chance that the four Australian sea-floor recorders would have heard, or not heard, each simulated impact at the times their recordings were running; the sound-coupling share is unknown and is averaged over 1e-6 to 1e-2. Orange: what the recorders say on their own if every latitude were equally likely beforehand. Dotted: adds the two deep-ocean stations with stand-in noise. The recorders were running about a third of the time, so most impacts could not have been heard at all. Dive probability 0.5. These are model predictions; no detection is claimed.

Technical note: Source posterior run J (74 seeds + northern supplement, p_core strata mix, seeds pooled by evidence), p = 0.5, per 00:19 option; module likelihood per row L = prod_r [(1 - P_D,r) + P_D,r m_r/beta_r], M1b, alpha = 0.05 per 55 s window, P_D from block 7 injection-recovery fits on own IMOS backgrounds (3376, 3315, 3274, 3275), TL block 2 arc grid (41-20 S; IMOS: RAM along the grid, imos_tl_ram_arc.csv, amendment A4), E = contact KE, tau = tau90 (EoF method 1), calibration error N(0, 10 dB), eta marginalised log-uniform [1e-6, 1e-2]; 'alone' = bin mean of exp(lnL) over bins with mass; KL(core||combined) D0' = 0.0082 nats. IMS stand-in = IMOS 3376 shaped to Blackman 2003 (digitised), +4.8 dB triad gain. PREDICTIVE, NOT CALIBRATED; caveats C1 (propagation slope misfit +10.4/+7.4 dB/oct at H01W/H08S) and C2 (coupling efficiency not calibrated, F-35A upper anchor 2.1e-4). Block 11 prereg 90d7181 + A4 e13a32e; module commits 90d7181, e13a32e; product run-j-gated-v2 (consistent table, built 2026-10-22T05:29:54 local).